

Anticoagulation challenges in cancer patients: prevention, detection, and treatment strategies

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Simona Chiusolo, Chelsea Kriesberg, Eric Oligino, Avirup Guha, Husam Abdel-Qadir, Joseph Carver, Stephen Casselli & Daniel Lenihan

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Anticoagulation Challenges in Cancer Patients: Prevention, Detection, and Treatment Strategies. S. Chiusolo^{1†}, C. Kriesberg^{2†}, E. Oligino³, A. Guha⁴, H. Abdel-Qadir⁵, J. Carver^{6,7}, S. Casselli⁷, D. Lenihan^{7,8}

¹ Department of Surgical, Medical and Molecular Pathology and Critical Area, Cardiology Division, University of Pisa, Pisa, Italy;

² Pediatric Cardio Oncology Department, Advocate Children's Heart Institute, Dempster St, Park Ridge, IL, USA;

³ Hartford Healthcare, Hartford, CT, USA;

⁴ Department of Medicine, Division of Cardiology, Medical College of Georgia at Augusta University, Augusta, GA, USA;

⁵ Women's College Hospital and University Health, Department of Medicine, University of Toronto, Toronto, Ontario, Canada;

⁶ University of Pennsylvania, Philadelphia, USA;

⁷ International Cardio-Oncology Society, Tampa, FL, USA;

⁸ Cardio-Oncology Center of Excellence and Heart Failure Clinic, Saint Francis Healthcare System, Cape Girardeau, MO, USA

[†] These authors contributed equally to this work.

Address for correspondence:

Daniel Lenihan, MD, FACC, FESC, FIC-OS

Cardio-Oncology Center of Excellence and Heart Failure Clinic

Saint Francis Healthcare System, 211 Saint Francis Drive, Cape Girardeau, MO 63703

cardio-oncologydoc@protonmail.com

Abstract

Cancer-associated thrombosis is one of the most common and serious complications in oncology patients, contributing significantly to morbidity and mortality. This prothrombotic state is driven by cancer-related factors, systemic inflammation, and treatment-induced endothelial injury. Anticoagulation management in these patients is challenging, as both thrombotic and bleeding risks must be carefully balanced, particularly in those with cardiovascular comorbidities. This review examines the complexities of anticoagulation in cancer-associated thrombosis through three clinical cases. Case 1 focuses on thrombosis associated with central venous catheters, exploring prevention strategies and therapeutic approaches. Case 2 discusses the management of venous thromboembolism in metastatic prostate cancer with concurrent cardiovascular disease, emphasizing the challenges of overlapping arterial and venous risks. Case 3 highlights the interplay between immune checkpoint inhibitors and postoperative atrial fibrillation in a patient with non-small cell lung cancer, exploring immune-mediated mechanisms and their implications for anticoagulation. The review also discusses advances in risk stratification tools, detection methods, and emerging therapies such as FXI inhibitors. Effective management of anticoagulation in cancer patients requires personalized strategies that consider cancer-specific risks, treatment-related factors, and comorbidities. Advances in biomarkers, predictive models, and innovative therapies hold promise for improving outcomes in this high-risk population.

Keywords: Cancer-associated thrombosis, anticoagulation, venous thromboembolism, direct oral anticoagulants, personalized medicine.

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1 Introduction

Cancer-associated thrombosis (CAT) is a frequent and serious complication in oncology patients, affecting up to 20% of individuals and ranking among the leading causes of non-cancer mortality[1-3]. This elevated thrombotic risk arises from a complex interplay of cancer biology, systemic inflammation, endothelial dysfunction, and treatment-induced vascular injury[3]. Tumor cells release procoagulants, including tissue factor and cancer procoagulant, which directly activate the clotting cascade. Concurrently, inflammatory cytokines such as interleukin-6 (IL-6) and tumor necrosis factor-alpha (TNF- α) exacerbate endothelial activation and platelet aggregation, fostering a prothrombotic environment[4, 5]. Cancer therapies significantly amplify these risks, further complicating anticoagulation management, inducing endothelial injury and favoring a hypercoagulable environment [6, 7]. Moreover, the presence of central venous catheters (CVCs) and other indwelling medical devices compounds this risk by disrupting vascular integrity and creating zones of stasis that promote thrombus formation. Catheter-related thrombosis (CRT) is particularly challenging, as it compromises treatment delivery and increases the risks of systemic embolism and infection[8]. For patients with pre-existing cardiovascular disease, the coexistence of cancer and atherosclerosis adds another layer of complexity, necessitating strategies that address both arterial and venous thrombotic risks[9]. Direct oral anticoagulants (DOACs) have emerged as a convenient and effective option for managing CAT, providing an alternative to low-molecular-weight heparins (LMWHs)[10, 11]. However, their use requires careful consideration of bleeding risks, drug interactions, and patient-specific factors, leaving room for the development of newer, safer therapeutic prospects[12]. This review explores the prevention, detection, and treatment of CAT through three illustrative clinical cases. These cases emphasize the intricate challenges of anticoagulation in oncology and highlight the need for multidisciplinary, patient-centered approaches.

2 Case vignettes

2.1 Case Vignette 1: Port-Associated Thrombosis

A 33-year-old female with a past medical history significant for hypertension, asthma, and depression presented with a palpable left breast mass diagnosed as invasive ductal carcinoma, estrogen receptor (ER) negative, human epidermal growth factor receptor 2 (HER2) positive, nuclear grade 3. She had no family or personal history of cardiac disease, cancer, hypercoagulability, or autoimmune disorders. Neoadjuvant chemotherapy with anthracycline-based therapy was initiated in November 2016, and her baseline echocardiogram demonstrated normal left ventricular ejection fraction (LVEF) of 55%. She completed four cycles of anthracycline-based chemotherapy in December 2016. A follow-up echocardiogram performed on January 6, 2017, demonstrated normal biventricular function but revealed a newly visualized right atrial mass (Figure 1A). Subsequent diagnostic imaging with computed tomography pulmonary embolism protocol (CT-PE) confirmed the mass in the right atrium (Figure 1B). Cardiac MRI was conducted for further tissue characterization and confirmed the diagnosis of a thrombus (Figure 1C). Treatment involved removal of the vascular access port on January 9, 2017, and initiation of therapeutic anticoagulation with low molecular weight heparin for three months. In May 2017, a new port was placed for the continuation of anti-HER2 targeted therapy. A follow-up echocardiogram in September 2017 identified another thrombus formation associated with this new port, which was subsequently removed. Chemotherapy was continued through peripheral intravenous access, and anticoagulation therapy with a direct oral anticoagulant was maintained until



Figure 1. Multipanel imaging assessment of intracardiac thrombus. 1A - Transthoracic echocardiography (TTE) showing a thrombus in the right atrium. 1B - Computed tomography angiography (CTA) confirming the presence of the right atrial mass. 1C - Cardiac magnetic resonance imaging (CMR) further characterizing the thrombus.

chemotherapy completion. The patient received chest wall radiation therapy in October 2017 and completed trastuzumab therapy in December 2017. A hypercoagulability evaluation performed in 2018 was negative, and no further thrombotic events occurred throughout her treatment course. See Clinical Vignette Discussion Boxes section for further insights.

2.2 Case Vignette 2: Cancer-Associated Thrombosis in a Patient with High Cardiovascular Risk

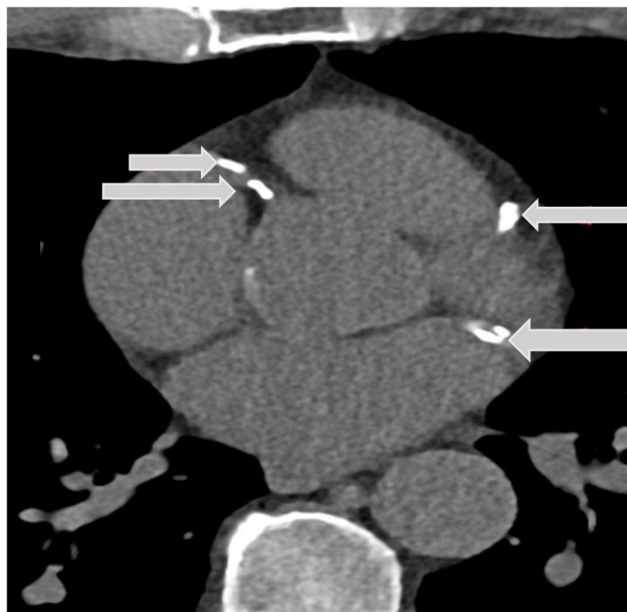


Figure 2: CT coronary angiography demonstrating extensive triple-vessel coronary artery disease

A 62-year-old man with a history of obesity and hypertension, managed with amlodipine, presented with an elevated PSA of 62 ng/mL. A subsequent biopsy revealed high-risk prostate cancer with Gleason-Piper splice findings. His hemoglobin was 9.8 g/dL, and his platelet count was 400,000/mm³. Following consultations with urologists and radiation oncologists, a PSMA PET scan was performed, which demonstrated two sites of metastatic disease, including involvement of the spine. He was started on a treatment regimen consisting of butyrolide, enzalutamide, and docetaxel. As part of institutional protocol, which mandates cardiovascular evaluation for all prostate cancer patients, the patient underwent a coronary calcium score evaluation, revealing a score of 2,420, with plaque distributed across multiple coronary arteries (Figure 2). Consequently, aspirin 81 mg daily and atorvastatin 40 mg were initiated for cardiovascular protection. Approximately one month later, the patient developed right leg swelling. A venous Doppler ultrasound revealed a positive finding for deep vein thrombosis (DVT), while a CT-PE scan was negative. After multi-disciplinary evaluation he was subsequently started on apixaban, and aspirin was discontinued. See Clinical

Vignette Discussion Boxes section for further insights.

2.3 Case Vignette 3: Cardiovascular Complication in a Patient on Immune Checkpoint Inhibitor Therapy

A 73-year-old man with a 30-pack-year smoking history, who quit two years before, presented with stage 3A non-small cell lung cancer (NSCLC). His past medical history was notable for hypertension, dyslipidemia, and mild chronic obstructive pulmonary disease. Despite the advanced cancer diagnosis, the patient maintained a high level of functionality, with an Eastern Cooperative Oncology Group (ECOG) performance status of 1. The patient underwent a thorough evaluation by a multidisciplinary team, including the surgical team, who deemed the tumor resectable despite the complexity of the case. Pathology confirmed a non-squamous subtype of NSCLC that was negative for epidermal growth factor receptor (EGFR) mutations and anaplastic lymphoma kinase (ALK) translocations but demonstrated a programmed death-ligand 1 (PD-L1) tumor proportion score of over 15%. This PD-L1 expression suggested potential benefit from immune checkpoint inhibitor therapy. The patient was treated with a neoadjuvant regimen consisting of nivolumab, pemetrexed, and cisplatin, administered every three weeks for three cycles. He tolerated the treatment well, experiencing mild nausea, pancytopenia, and a rash, which resolved with a short course of prednisone. No significant immune-related complications were observed, and cardiac biomarkers were monitored only when clinically indicated. Following completion of chemotherapy, a post-treatment CT scan showed no disease progression. The patient proceeded to surgery four weeks later. Surgical resection was performed without complications. On postoperative day two, the patient developed new-onset chest tightness and dyspnea. His vital signs were stable except for tachycardia and an irregular heartbeat. Clinical evaluation and investigations revealed new-onset atrial fibrillation (AF). An echocardiogram showed a left atrium at the upper limit of normal size, without significant valvular disease or myocardial dysfunction. Troponin levels were minimally elevated, but there were no other signs of myocardial injury. The patient reverted to sinus rhythm spontaneously within 20 hours, without the need for antiarrhythmic drugs. He remained in sinus rhythm for the rest of his hospital stay, and his discharge on day eight was

uneventful, without initiation of anticoagulation therapy but with a plan for close follow-up monitoring. At his three-month follow-up, the patient reported a transient episode of right-sided weakness and speech difficulty that resolved within minutes. He was evaluated in the emergency department, where neurological examination was unremarkable, but brain MRI confirmed a small acute ischemic lesion consistent with embolic stroke. An electrocardiogram performed on admission unexpectedly revealed atrial fibrillation, despite the patient being asymptomatic. To assess for possible large-vessel involvement, a cerebral CT angiography was performed and excluded stenosis or occlusion of the cervical or intracranial arteries, strengthening the suspicion of a cardioembolic mechanism. Given the recurrence of AF and the new ischemic event, long-term anticoagulation therapy was initiated. The patient was subsequently enrolled in a structured cardio-oncology follow-up program focusing on rhythm monitoring, thromboembolic prevention, and surveillance for potential cancer- and therapy-related cardiovascular complications. See Clinical Vignette Discussion Boxes section for further insights.

3. Prevention of Cancer-Associated Thrombosis

Effective prevention of CAT is vital for minimizing morbidity and mortality in oncology patients, especially those with heightened cardiovascular risk. Achieving this requires a comprehensive approach that combines advanced risk stratification tools with both pharmacologic and non-pharmacologic interventions. These strategies must be tailored to the individual patient, accounting for cancer type, treatment modalities, comorbidities, and device-related risks. By integrating evidence-based practices and personalized care plans, clinicians can better address the complex interplay of systemic and localized thrombotic factors in this high-risk population.

3.1 Risk Stratification Tools

Risk stratification for thrombotic events in cancer patients requires distinguishing between venous and arterial mechanisms, as these entities arise from different pathophysiologic pathways and demand tailored assessment strategies.

Accurate risk stratification is essential for effective prevention of venous thromboembolism (VTE), allowing clinicians to identify high-risk patients and tailor prophylaxis (Table 1). The Khorana Score remains widely used due to its simplicity but has moderate predictive power (c-statistic: 0.65–0.70) and omits key variables such as cardiovascular comorbidities and treatment details [13, 14]. To address these gaps, newer models like the EHR-CAT Risk Assessment Model (RAM) have been developed. Derived from nearly 90,000 cancer patients and validated in over 2.1 million individuals, EHR-CAT incorporates cancer-specific, clinical, and demographic factors, achieving superior predictive performance (c-statistic ~0.71)[15, 16]. For specific patient populations, the Thrombosis Lymphoma (ThroLy) Score has emerged as a tailored risk assessment model for patients with lymphoma, addressing the unique thrombotic risk factors associated with this malignancy[17, 18]. By incorporating disease-specific variables, the ThroLy Score provides a more nuanced risk assessment compared to generalized tools, facilitating improved decision-making for prophylaxis in this subgroup[18].

Risk stratification for arterial thrombotic events (ATE) in cancer patients requires a different approach, as mechanisms and risk factors are distinct from those driving venous thrombosis. Cancer patients frequently have coexisting cardiovascular comorbidities that increase baseline arterial risk, which may be further amplified by cardiotoxic therapies such as platinum agents, VEGF inhibitors, and hormonal treatments. The European Society of Cardiology (ESC) and International Cardio-Oncology Society (ICOS) guidelines recommend structured assessments using tools like SCORE2 or coronary calcium scoring[11].

Incorporating these assessments into CAT prevention strategies enables clinicians to address both venous and arterial risks comprehensively. Emerging tools and innovations hold promise for advancing CAT prevention. Predictive models like ONCOTHROMB, which integrate clinical, genetic, and biomarker data, may provide more precise risk stratification[19].

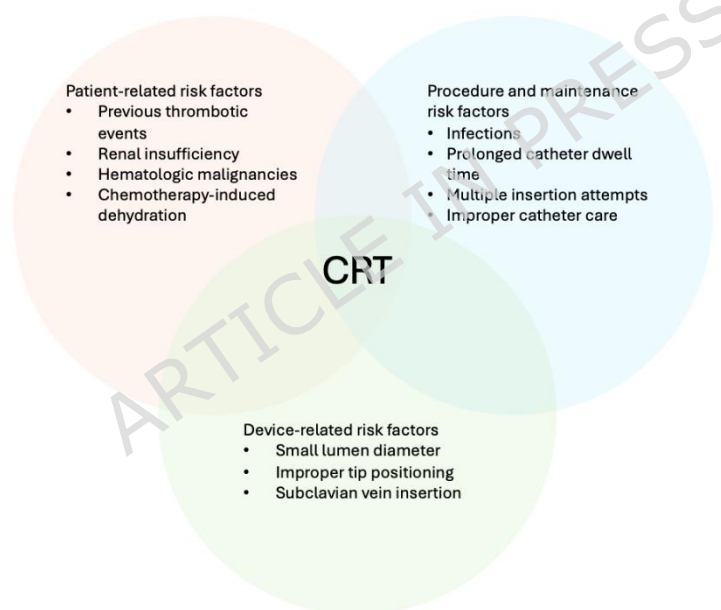
Score	Target Population	Main Variables	Risk Stratification	Strengths	Limitations
Khorana Score	Ambulatory cancer patients starting chemotherapy	Cancer type, Platelet count $\geq 350,000/\mu\text{L}$, Hb $< 10 \text{ g/dL}$ or ESA use, WBC $> 11,000/\mu\text{L}$, BMI $\geq 35 \text{ kg/m}^2$	0 = Low, 1–2 = Intermediate, ≥ 3 = High	Widely validated, easy to calculate, endorsed by guidelines (ASCO, ISTH)	Limited sensitivity; underestimates risk in some cancers (e.g., lung, GI); not validated in Multiple Myeloma

EHR-CAT	Electronic Health Record-based; broad cancer cohorts	Cancer site, Previous VTE, Age, BMI, Hb, Platelets, WBC, Other labs (creatinine, albumin)	Continuous score; often stratified into tertiles/quartiles	Real-world applicability, uses EHR data, dynamic risk estimation	Requires access to comprehensive EHR data; less familiar to clinicians
ThroLy Score	Lymphoma patients	Tumor bulk, Mediastinal involvement, Platelets $\geq 350,000/\mu\text{L}$, Hb $< 10 \text{ g/dL}$, ECOG ≥ 2 , Prior thrombosis, Extranodal involvement	0–1 = Low, 2–3 = Intermediate, ≥ 4 = High	Specific to lymphoma, good internal validation	Limited to lymphoma, not applicable to solid tumors

Table 1 Comparison of Risk Stratification Tools for Cancer-Associated Thrombosis. This table summarizes the key features of three widely used risk assessment models for venous thromboembolism in cancer patients: the Khorana Score, EHR-CAT, and ThroLy Score. BMI – Body Mass Index, ECOG – Eastern Cooperative Oncology Group, EHR-CAT – Electronic Health Record–Cancer-Associated Thrombosis, ESA – Erythropoiesis-Stimulating Agent, Hb – Hemoglobin, VTE – Venous Thromboembolism, WBC – White Blood Cell (count)

3.2 Catheter-Related Thrombosis: A Specific Subset

Catheter-related thrombosis (CRT) is a major contributor to CAT, particularly in cancer patients requiring central venous access devices (CVADs) for chemotherapy, blood transfusions, or parenteral nutrition[8, 20]. Several risk factors may contribute to the development of CRT (Fig. 3). CVADs, including peripherally inserted central catheters (PICCs), central venous catheters (CVCs), and implanted ports, are indispensable for treatment but significantly increase thrombotic risk. Studies show that PICCs are associated with a higher CRT risk compared to implanted ports, likely due to venous stasis and reduced luminal diameter[8]. Improper catheter tip positioning, particularly outside the superior vena cava or atrial-caval junction, doubles CRT risk, while



subclavian vein insertion further exacerbates the likelihood of thrombosis[8]. Patient-related factors such as prior thrombotic events, renal insufficiency, and hematologic malignancies like multiple myeloma also increase CRT risk[8, 21]. Infections associated with CVADs, dehydration due to chemotherapy-induced gastrointestinal complications, and prolonged catheter dwell times are additional contributors. Preventing CRT involves optimizing catheter selection, insertion techniques, and maintenance protocols. Implanted ports should be prioritized over PICCs for long-term use[21]. Ultrasound-guided insertion and fluoroscopic confirmation of catheter tip placement at the atrial-caval junction are critical for reducing mechanical and positional risks. Routine flushing with saline or heparinized solutions and vigilant monitoring for early signs of infection or

Figure 1 Risk Factors for Catheter-Related Thrombosis (CRT). This diagram illustrates key risk factors for CRT, categorized into device-related factors, patient-related factors, and procedure & maintenance factors. Preventative strategies should focus on optimal catheter selection, proper insertion techniques, and rigorous maintenance protocols to reduce thrombotic complications.

occlusion further mitigate CRT risk[20, 21].

3.4 Pharmacologic and Non-Pharmacologic Approaches

Given the distinct mechanisms and clinical consequences of venous and arterial thrombosis in oncology, preventive strategies should be addressed separately for VTE and ATE.

Venous thromboprophylaxis is a cornerstone of CAT prevention, and current guidelines emphasize an individualized approach based on patient-specific thrombotic and bleeding risks (Table 2). The American Society of Hematology guidelines emphasize a tailored approach based on the individual risk profile of cancer patients. For ambulatory patients at low risk of VTE, routine thromboprophylaxis is not recommended to avoid unnecessary bleeding risks[22]. However, in hospitalized medical patients with cancer, thromboprophylaxis is advised when the risk of VTE outweighs the risk of bleeding, with LMWH or fondaparinux being preferred agents. Surgical cancer patients, particularly those undergoing major surgery, benefit from perioperative thromboprophylaxis using LMWH or fondaparinux. For high-risk ambulatory patients, such as those with a Khorana Score of 2 or higher and undergoing systemic therapy, LMWH or DOACs such as apixaban or rivaroxaban are effective options[22]. The 2022 International Clinical Practice Guidelines from the International Initiative on Thrombosis and Cancer (ITAC) align with many of ASH recommendations while offering additional considerations[10]. ITAC strongly recommends primary thromboprophylaxis with LMWHs or DOACs in ambulatory patients with locally advanced or metastatic pancreatic cancer receiving systemic therapy, provided the bleeding risk is low. Additionally, extended postoperative thromboprophylaxis lasting four weeks with LMWHs is advised for patients undergoing major abdominopelvic surgery who do not have a high risk of bleeding[10]. For CRT, pharmacologic prophylaxis with LMWH or DOACs may be considered in high-risk patients, such as those with a history of thrombosis or hematologic malignancies[8]. However, given the delicate balance between preventing thrombotic events and minimizing bleeding complications, individualized risk assessment is critical[23]. Non-pharmacologic strategies also play an important role, particularly for patients at elevated bleeding risk. Mechanical interventions, such as intermittent pneumatic compression devices and graduated compression stockings, are effective in preventing venous stasis, especially in hospitalized or post-operative patients[24]. Lifestyle modifications remain a key adjunct in thromboprophylaxis. Smoking cessation, regular physical activity, and maintaining adequate hydration are fundamental measures that support vascular health and reduce thrombotic risk. Additionally, patient education is vital to ensuring adherence to prescribed therapies and fostering early recognition of VTE symptoms, which can improve outcomes and prevent complications[10].

Category	Recommendation
Ambulatory Cancer Patients (Low VTE Risk)	Routine thromboprophylaxis NOT recommended (to avoid bleeding risks).
Ambulatory Cancer Patients (High VTE Risk, Khorana Score ≥ 2)	LMWH or DOACs (apixaban, rivaroxaban) recommended for thromboprophylaxis.
Hospitalized Cancer Patients	Thromboprophylaxis recommended if VTE risk outweighs bleeding risk (preferred agents: LMWH, fondaparinux).
Surgical Cancer Patients (Perioperative)	Perioperative thromboprophylaxis recommended (preferred agents: LMWH, fondaparinux).
Surgical Cancer Patients (Postoperative)	Extended postoperative thromboprophylaxis (4 weeks) with LMWH recommended if low bleeding risk.
Patients with Locally Advanced or Metastatic Pancreatic Cancer Receiving Systemic Therapy	Primary thromboprophylaxis with LMWH or DOACs strongly recommended if bleeding risk is low.
Patients with CRT (Catheter-Related Thrombosis)	Pharmacologic prophylaxis (LMWH or DOACs) may be considered in high-risk patients (e.g., history of thrombosis, hematologic malignancies).

Table 2 This table summarizes the key recommendations for thromboprophylaxis in different cancer patient populations based on ASH and ITAC guidelines. Pharmacologic strategies, including LMWH and DOACs (apixaban, rivaroxaban), are recommended for high-risk ambulatory patients, hospitalized cancer patients, and those undergoing major surgery. Extended postoperative prophylaxis is advised for surgical patients at low bleeding risk. ASH – American Society of Hematology. CRT – Catheter-Related Thrombosis; DOACs – Direct Oral Anticoagulants; ITAC – International Initiative on Thrombosis and Cancer; LMWH – Low-Molecular-Weight Heparin; VTE – Venous Thromboembolism.

Arterial thrombosis prevention in cancer patients focuses on optimizing cardiovascular risk factors and mitigating therapy-related vascular toxicity. Unlike VTE prophylaxis, routine pharmacologic prevention of arterial thrombotic events is not recommended in unselected cancer populations [11]. Instead, arterial risk mitigation relies on structured cardiovascular assessment, including lipid profiling, blood pressure control, glycemic

optimization, and smoking cessation. Guidelines from ESC and ICOS emphasize the use of structured tools such as SCORE2 and coronary calcium scoring (CAC) to estimate baseline arterial risk and guide primary prevention strategies [11]. Patients with established atherosclerotic cardiovascular disease or high CAC scores benefit from statin therapy and, when indicated, aspirin for secondary prevention, provided bleeding risk is acceptable. Cancer therapies known to increase arterial risk—including VEGF inhibitors, platinum-based regimens, and fluoropyrimidines—require proactive cardiovascular optimization, blood pressure control, and prompt management of emerging symptoms suggestive of ischemia. Unlike VTE, routine anticoagulation or antiplatelet therapy is not recommended solely for ATE prevention in cancer patients unless standard cardiology indications are present (e.g., prior myocardial infarction, established CAD, PAD).

4. CAT Detection

4.1 Detection of VTE

The detection of venous thromboembolism (VTE) in cancer patients requires an individualized and structured diagnostic strategy. Doppler ultrasound is the first-line modality for deep vein thrombosis (DVT) due to its accessibility, safety, and diagnostic accuracy [25]. CT pulmonary angiography (CTPA) remains the gold standard for the diagnosis of pulmonary embolism (PE), providing high sensitivity and specificity [25]. In selected scenarios, additional imaging tools can support VTE diagnosis. FDG-PET/CT may incidentally reveal metabolically active thrombi during oncologic staging, although its limited specificity necessitates confirmation with Doppler ultrasound or contrast-enhanced CT to avoid false positives [26]. Contrast-enhanced ultrasound (CEUS) offers real-time, non-nephrotoxic imaging and is particularly useful in challenging venous territories or in portal vein thrombosis associated with hepatocellular carcinoma, where it can differentiate bland from neoplastic thrombi through characteristic enhancement patterns [27-29]. Biomarkers complement imaging in VTE assessment. D-dimer is widely used due to its high negative predictive value; however, nonspecific elevations related to tumor burden, inflammation, or advanced age may reduce specificity in oncology patients. Age-adjusted thresholds improve diagnostic performance in older individuals with comorbidities [30-32]. Soluble P-selectin, reflecting platelet and endothelial activation, has shown promise as a biomarker of hypercoagulability, although standardized cut-offs and large-scale validation are still lacking [33, 34]. Advances in molecular profiling have identified circulating microRNAs (miRNAs) involved in endothelial activation and coagulation, which may serve as future non-invasive biomarkers for VTE detection [35].

4.2 Detection of ATE

Detection of ATE in cancer patients relies primarily on cardiovascular imaging and biomarkers of myocardial injury, distinguishing acute ischemic events from therapy-related cardiotoxicity. Coronary CT and CT angiography (CCTA) are essential in detecting coronary artery disease, particularly in patients exposed to pro-thrombotic chemotherapy agents like cisplatin or VEGF inhibitors (VEGFi) [11]. Transthoracic echocardiography (TTE) plays a critical role in evaluating cardiac function, detecting cardiac and vena cava thrombi, and monitoring structural changes over time. It is especially useful in patients with central venous devices or compressive tumors. Contrast-enhanced echocardiographic techniques further improve thrombus visualization [39, 40]. Additionally, transesophageal echocardiography (TEE), though more invasive, complements TTE by providing superior resolution and detailed imaging of the vena cava, particularly in cases of suspected thrombi in anatomically challenging regions [41-43]. Emerging evidence suggests that combining these modalities with advanced techniques, such as three-dimensional (3D) imaging and strain analysis, can offer a comprehensive assessment of thrombus burden and vascular integrity, supporting more informed clinical decision-making in high-risk patients [30, 44, 45]. Biomarkers in ATE detection focus on identifying myocardial injury. High-sensitivity troponins are the cornerstone for diagnosing ischemia or cancer therapy-related myocardial damage. Rising or dynamic troponin elevations provide strong diagnostic value in differentiating acute coronary syndromes from inflammatory or cardiotoxic etiologies. In contrast, systemic inflammatory markers such as high-sensitivity C-reactive protein (hsCRP) and atherosclerotic risk markers such as lipoprotein(a) [Lp(a)] are not diagnostic for acute ATE but contribute to risk stratification, helping identify patients with heightened long-term arterial vulnerability—particularly in those receiving cardiotoxic therapies [36, 37, 38]. Advances in molecular profiling have led to multi-biomarker panels integrating hsCRP, Lp(a), troponins, and traditional cardiovascular markers to enhance thrombotic risk stratification in cancer patients [37]. Future studies are needed to validate these panels and assess their predictive accuracy in diverse cancer populations.

5. Cancer Therapy-Associated Thromboembolic Complications

Thromboembolic events are among the most significant complications associated with cancer therapies, driven by a multifaceted interplay between treatment modalities and patient-specific risk factors[36]. Both venous and arterial thrombotic events can arise due to the pro-thrombotic and endothelial-disrupting effects of systemic therapies, radiotherapy, and targeted agents, compounding the already elevated baseline risk in cancer patients[36]. Chemotherapy agents, particularly platinum-based therapies like cisplatin, are well-known to increase the risk of both VTE and ATE[37, 38]. Cisplatin induces endothelial damage, promotes platelet activation, and alters fibrinolysis, leading to a hypercoagulable state. Studies have consistently shown that cisplatin-based regimens double the risk of thromboembolic complications compared to non-platinum agents[38]. Taxanes, commonly used in breast and ovarian cancers, also increase thrombotic risk through vascular inflammation and microvascular damage[39]. Hormonal therapies, such as tamoxifen and aromatase inhibitors, are widely used in hormone receptor-positive breast cancer. Tamoxifen, in particular, is associated with a 2-3 fold increased risk of VTE due to its estrogenic effects on coagulation[40, 41]. In contrast, aromatase inhibitors appear to have a lower thrombotic risk but are not entirely devoid of vascular effects, particularly in patients with pre-existing cardiovascular disease[42]. Chimeric antigen receptor T-cell (CAR-T) therapies, while highly effective in hematologic malignancies, have also been associated with coagulopathy and thrombotic events, primarily through mechanisms related to cytokine release syndrome (CRS) and endothelial activation. Severe CRS can trigger a cascade of inflammatory mediators, leading to disseminated intravascular coagulation (DIC), thrombotic microangiopathy, and elevated VTE risk[43, 44]. Monitoring coagulation parameters during CAR-T therapy is therefore critical to mitigate these potentially life-threatening complications[44]. Immune checkpoint inhibitors (ICIs), such as nivolumab and pembrolizumab, have revolutionized cancer therapy but introduce unique thromboembolic challenges[40, 45]. ICIs can trigger immune-related adverse events, including myocarditis, vasculitis, and endothelial dysfunction, which predispose patients to both VTE and ATE[46]. A recent meta-analysis identified atrial fibrillation as the most frequently observed complication of ICIs, occurring in 12% of cases. This highlights the propensity of ICIs to disrupt cardiac electrophysiology. More severe outcomes, including cardiac arrest and cardiac failure, were reported in 6% of cases, with three fatalities documented. The pooled analysis demonstrated that ICI therapy significantly increases the relative risk of cardiotoxicity compared to control groups. These findings underscore that, although less common than other adverse effects, the cardiotoxic potential of ICIs is clinically significant and potentially life-threatening. Agents targeting the VEGF pathway, such as bevacizumab, are strongly linked to arterial thrombotic events, including myocardial infarction and stroke. By inhibiting angiogenesis, these therapies compromise vascular integrity and endothelial repair, creating a pro-thrombotic state[47]. They also elevate the risk of hypertension, a key driver of arterial thrombosis, necessitating close cardiovascular monitoring.[48] Radiotherapy further exacerbates thrombotic risks through vascular damage and inflammation, particularly in patients receiving chest or pelvic radiation, which increases arterial and venous thrombosis risk. Targeted therapies like tyrosine kinase inhibitors (e.g., ponatinib) add to this burden by disrupting platelet and endothelial function[49]. Careful risk assessment and proactive management are essential, especially for patients with pre-existing thrombotic conditions[11].

Therapy Type	Thromboembolic Risk
Platinum-Based Chemotherapy (e.g., Cisplatin)	Doubles the risk of VTE and ATE; induces endothelial damage, platelet activation, and fibrinolysis alteration.
Taxanes (e.g., Paclitaxel, Docetaxel)	Increases thrombotic risk through vascular inflammation and microvascular damage.
Hormonal Therapy (e.g., Tamoxifen, Aromatase Inhibitors)	Tamoxifen: 2-3x increased VTE risk due to estrogenic effects; Aromatase inhibitors: Lower risk but vascular effects in CVD patients.
CAR-T therapies (e.g., axicabtagene ciloleucel)	Increased risk of disseminated intravascular coagulation, thrombotic microangiopathy, and VTE
Immune Checkpoint Inhibitors (e.g., Nivolumab, Pembrolizumab)	Triggers immune-related adverse events (myocarditis, vasculitis, endothelial dysfunction); increases VTE and ATE risk.
VEGF Pathway Inhibitors (e.g., Bevacizumab)	Strongly linked to arterial thrombotic events (MI, stroke); compromises vascular integrity; increases hypertension risk.
Radiotherapy	Exacerbates thrombotic risks through vascular damage and inflammation; chest/pelvic radiation increases both VTE and ATE risk.
Tyrosine Kinase Inhibitors (e.g., Ponatinib)	Disrupts platelet and endothelial function; increases thrombosis risk.

Table 1 Thromboembolic Risk of Cancer Therapies. This table outlines the thrombotic risks of various cancer treatments. ATE – Arterial Thromboembolism, CAR-T – Chimeric antigen receptor T-cell, CVD – Cardiovascular Disease, ICI – Immune Checkpoint Inhibitor, MI – Myocardial Infarction, VTE – Venous Thromboembolism, VEGF – Vascular Endothelial Growth Factor.

Initial Diagnosis & Risk Stratification

- Thrombotic risk (e.g., VTE, arterial events)
- Bleeding risk (e.g., GI malignancies, prior major bleeding)
- Cancer type considerations

Choice of Anticoagulant Based on Risk Factors

- DOACs (e.g., Apixaban, Rivaroxaban) – Preferred in most cases except for high bleeding risk
- LMWHs (e.g., Enoxaparin) – Preferred for GI cancers or patients with high bleeding risk
- VKAs (Warfarin) – Generally avoided due to variability in effect and interactions

Special Considerations in Cancer Patients with Cardiovascular Disease

- Dual antiplatelet therapy (DAPT) considerations in ACS/PCI patients
- Strategies to **minimize bleeding risk** (e.g., early de-escalation of antiplatelet therapy, transitioning to single therapy)

6. CAT Treatment

Cancer-associated thrombosis (CAT) requires a careful balance between preventing recurrent thrombosis and minimizing bleeding risk, given the complex clinical profile of oncology patients. Management follows a

Figure 2 Decision pathway for cancer-associated thrombosis (CAT) management.

structured pathway involving comprehensive risk assessment, individualized anticoagulant selection, and evaluation of concomitant therapies that may amplify bleeding risk (Figure 5)[23].

DOACs, with fixed dosing and oral administration, have become essential in the treatment and prevention of VTE, including DVT and PE, simplifying adherence and offering a favorable efficacy profile in this high-risk population. Current guidelines, including those from the ASH and ESC, recommend DOACs such as apixaban and rivaroxaban as first-line options for treating VTE in cancer patients without high bleeding risk or gastrointestinal malignancies[11, 22]. The SELECT-D trial demonstrated rivaroxaban's effectiveness for treating VTE in cancer patients[50], while the AVERT trial highlighted the role of apixaban in VTE prevention for high-risk ambulatory cancer patients[51]. Further confirming the role of DOACs, the CARAVAGGIO trial in 2020 established the safety and efficacy of apixaban as an alternative to LMWHs for treating VTE in patients with active cancer[52]. In cases where there is a significant bleeding risk or concerns about drug interactions, LMWHs remain preferred due to their established safety profile and lower interaction potential[10, 22]. LMWHs, such as enoxaparin, offer advantages especially for patients with gastrointestinal cancers or conditions that complicate DOAC use, as they do not rely on gastrointestinal absorption and provide more predictable pharmacokinetics in certain cancer populations[10]. While LMWH and DOACs are the mainstays of VTE treatment in cancer patients, vitamin K antagonists (VKAs) like warfarin are generally avoided[10]. VKAs present challenges due to variable interactions with cancer therapies, dietary restrictions, and a need for frequent monitoring to maintain therapeutic levels, which complicates their use in this population. Additionally, VKAs require dose adjustments influenced by factors like drug-drug interactions, common in cancer treatment, making them less practical and less safe in cancer-associated VTE[53, 54]. Dose adjustments are critical in the use of both LMWHs and DOACs, especially in patients with renal impairment, as renal function significantly affects drug clearance. Careful monitoring and tailored dosing of these anticoagulants are essential to maximize efficacy while minimizing bleeding risk in cancer patients managing VTE[12]. The long-term management of CAT remains an evolving field. The API-CAT

trial, a randomized double-blind noninferiority study, evaluated reduced-dose apixaban (2.5 mg BID) versus full-dose therapy (5 mg BID) for extended anticoagulation beyond six months. Reduced-dose apixaban demonstrated comparable efficacy with a more favorable bleeding profile, supporting its use as an extended-therapy option in patients with active cancer at increased bleeding risk [55].

The management of ATE in cancer patients focuses on secondary prevention of atherosclerotic events and the optimization of cardiovascular risk, rather than routine anticoagulation. Patients with significant coronary plaque burden—identified through elevated coronary calcium scores or a history of chronic coronary syndrome—often require antiplatelet therapy, typically aspirin, as part of standard secondary prevention [11]. Statin therapy is also recommended for patients with established cardiovascular disease or high atherosclerotic risk to stabilize plaque and reduce both arterial and venous thrombotic risks[11]. For patients undergoing recent coronary intervention, particularly following an acute coronary syndrome (ACS), a short course of dual antiplatelet therapy with aspirin and clopidogrel is typically recommended for at least one-three month, after which transitioning to single antiplatelet therapy helps to balance thrombotic protection with bleeding risk[56]. When considering the use of more potent P2Y₁₂ inhibitors, such as ticagrelor or prasugrel, it's important to carefully evaluate both bleeding and thrombotic risks, as these agents may increase bleeding potential in high-risk patients[56].

In cancer patients who require both antiplatelet and anticoagulant therapy due to AF or VTE, the primary goal is to minimize bleeding risk while maintaining adequate thrombotic protection. A critical strategy in this setting is the early de-escalation of antiplatelet therapy[57]. Treatment generally begins with triple therapy—aspirin, clopidogrel, and anticoagulation—but is limited to a maximum of 7 days post-intervention to reduce bleeding risk[56]. Following this initial period, therapy is typically streamlined to a single antiplatelet agent, usually clopidogrel, combined with anticoagulation. This approach provides ongoing protection against both VTE and arterial events while managing bleeding risk effectively. For patients requiring anticoagulation without recent ACS or percutaneous coronary intervention, anticoagulation alone may suffice, with antiplatelet therapy considered only very selected cases with significant plaque burden at low bleeding risk[56, 58, 59].

Innovative treatments are expanding therapeutic options for managing CAT, particularly for patients at high risk of bleeding complications. Among these, FXI inhibitors show promise as alternatives to traditional anticoagulants by targeting later stages of the coagulation cascade[60]. This targeted approach potentially reduces bleeding risks while providing effective clot prevention, making FXI inhibitors particularly suited for cancer patients who often present complex thrombotic and bleeding profiles[46]. Several types of FXI inhibitors, including antisense oligonucleotides (ASOs), monoclonal antibodies (mAbs), and small molecules, are showing promise as safer anticoagulants. In orthopedic surgery, such as total knee arthroplasty, FXI inhibitors have effectively reduced VTE risk with fewer bleeding complications. For instance, the ASO IONIS-FXIRX significantly outperformed enoxaparin in a phase 2 trial[61], while monoclonal antibodies like osocimab and abelacimab and the small molecule inhibitor milvexian demonstrated comparable efficacy to enoxaparin with reduced bleeding risks[62-64]. In cancer patients, emerging data suggest FXI inhibitors like abelacimab and xisomab 3G3 may provide additional protection against catheter-associated thrombosis during chemotherapy, maintaining an acceptable safety profile[64]. These emerging treatments signal a shift toward safer and more targeted anticoagulation options, with FXI inhibitors potentially transforming the management of CAT by providing tailored strategies that balance thrombotic protection and bleeding risks[46]. The expanding research in FXI inhibitors and other novel agents indicates a promising future for integrated thrombotic management in cancer patients, particularly for those with concurrent cardiovascular disease or complex thrombotic profiles.

7. Discussion

Anticoagulant therapy should be adapted to cancer type and bleeding risk. CAT represents a significant challenge in oncology, reflecting the intricate interplay between cancer biology, systemic inflammation, and treatment-induced endothelial injury. This review highlights the complexities of CAT prevention, detection, and management, underscoring the need for individualized approaches tailored to patient-specific risk factors, cancer subtypes, and therapeutic regimens. Recent advancements in risk stratification tools demonstrate progress in identifying high-risk patients[16]. These tools, coupled with emerging biomarkers, enable more precise risk assessments and early detection of both venous and arterial thrombotic events. Innovations in imaging modalities further enhance diagnostic accuracy by offering non-invasive, real-time evaluations of thrombotic burden and vascular integrity[3]. However, the management of CAT still requires a delicate balance between effective anticoagulation and the mitigation of bleeding risks. The integration of cardiovascular care into oncology is crucial for optimizing outcomes in patients with CAT. Multidisciplinary collaboration among

oncologists, cardiologists, and hematologists is essential to navigate the complexities of CAT management. Future research should focus on validating these innovations across diverse cancer populations and exploring personalized approaches to mitigate the burden of CAT.

7.1 Vignette Discussion Boxes

Clinical Vignette Discussion Box 1: CRT in Cancer Patients

Mini Discussion: This case illustrates the importance of a proactive approach to CRT in cancer patients. Risk factors vary depending on cancer type, treatment regimen, and patient comorbidities. A structured strategy involving risk stratification, tailored prophylaxis, appropriate imaging, and individualized anticoagulation is essential to optimize outcomes and minimize complications.

Learning Points:

- CRT is a common but preventable complication in oncology.
- Risk assessment tools should be supplemented with clinical judgment.
- Diagnostic escalation to CT/MRI may be required to differentiate thrombus from other masses.

Clinical Vignette Discussion Box 2: Managing Dual Thrombotic Risk in Metastatic Prostate Cancer

Mini Discussion: This case underscores the need for balancing venous and arterial thrombotic risks in cancer patients, especially those on potential pro-thrombotic agents. Initial cardiovascular prevention strategies must evolve as new thrombotic events occur, requiring flexible and patient-specific anticoagulation choices that account for bleeding risk and drug-drug interactions.

Learning points:

- Cancer patients may face both venous and arterial thrombotic risks, particularly when treated with agents like androgen deprivation therapy, VEGF inhibitors, or hormonal agents (e.g., enzalutamide).
- Initial antiplatelet therapy for arterial risk (e.g., aspirin + statin) may need to be adjusted once VTE is diagnosed.
- Avoidance of dual-pathway antithrombotic therapy (e.g., antiplatelet + anticoagulant) is often necessary to reduce bleeding risk, especially in patients with elevated bleeding susceptibility.
- DOAC selection should consider drug-drug interactions.
- Ongoing clinical reassessment and imaging are essential, as thrombotic risk is dynamic and influenced by cancer progression and treatment

Clinical Vignette Discussion Box 3: Atrial Fibrillation in NSCLC on ICI Therapy

Mini Discussion: This case illustrates the importance of distinguishing postoperative atrial fibrillation (POAF) from incident or recurrent AF in cancer patients. The initial AF episode—transient, self-limited, and occurring in the immediate postoperative period—was consistent with POAF, for which short-term anticoagulation is often unnecessary. However, the later embolic stroke and documentation of AF during follow-up highlight how transient POAF can coexist with, or evolve into, clinically meaningful AF requiring long-term management. In oncology patients, especially those with cardiovascular comorbidities or receiving therapies such as immune checkpoint inhibitors, structured rhythm surveillance and dynamic reassessment of thromboembolic risk are essential to guide safe and effective anticoagulation decisions.

Learning points:

- POAF and later-onset AF represent different clinical entities: POAF is often transient, whereas recurrent or incident AF weeks to months later carries a higher long-term thromboembolic risk.
- AF in cancer patients requires a broad etiologic assessment, including perioperative triggers, systemic inflammation, structural heart disease, and, when relevant, immune-related adverse events.
- A comprehensive diagnostic workup—including echocardiography, cardiac biomarkers, and cerebrovascular imaging—helps differentiate POAF from AF associated with underlying cardiomyopathy or cardioembolism.
- Short-lived POAF (<48 hours) rarely requires anticoagulation, but the emergence of recurrent AF or a thromboembolic event mandates re-evaluation and often initiation of long-term anticoagulation.
- Continuous or periodic rhythm monitoring is critical, as subclinical AF is common and may change management in high-risk oncology patients.
- Anticoagulation decisions must be individualized, balancing stroke prevention against bleeding risk and adapting dynamically as the patient's clinical status evolves.

8. Conclusions

CAT remains a significant contributor to morbidity and mortality in oncology patients. Its management demands a nuanced approach, balancing thrombotic and bleeding risks while integrating individualized risk stratification and emerging therapies. Advances in biomarkers, predictive models, and imaging modalities have improved our ability to identify and monitor high-risk patients, enabling earlier and more targeted interventions. The interplay of cancer treatments and thrombotic risks necessitates a multidisciplinary approach, emphasizing collaboration between oncologists, cardiologists, and hematologists to navigate the complexities of CAT. Future research should focus on validating innovative therapies, refining risk stratification tools, and addressing gaps in current guidelines to ensure safer and more effective management. Ultimately, the successful management of CAT requires continuous evolution in clinical practice, bridging oncology and cardiovascular care to improve patient outcomes and quality of life.

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